

Project Title

The Human Error Simulator – A Gamified Behavioral Risk Analysis Platform

What We Are Building

We are building a **web-based simulation platform** that recreates realistic social engineering attacks inside a controlled, desktop-like environment.

But this is NOT just a phishing training tool.

It is a:

Behavioral analytics system that measures how humans react under social engineering pressure.

The core objective is:

- Simulate realistic attack scenarios
- Observe user decisions
- Capture behavioral signals
- Convert those signals into measurable risk insights

System Overview (High-Level)

The application consists of five major layers:

1. User Authentication Layer
2. Desktop Simulation Engine
3. Interaction Tracking Engine
4. Risk Evaluation Engine
5. Analytics Dashboard

AI is used only to enhance realism and explanation — not to control risk decisions.

1. User Authentication Layer

Users log in using Firebase authentication.

Each user has:

- Their own simulation history
- Their own behavioral profile
- Their own vulnerability metrics

This makes the system personalized.

2. Desktop Simulation Environment

Instead of static quizzes, the user enters a simulated Windows-like desktop environment.

Inside this environment:

- A Mail application exists
- A Browser application exists
- Windows can be minimized, maximized, dragged
- Multiple windows can stay open simultaneously

This design choice is intentional.

Why?

Because social engineering happens in real workflows — not in multiple-choice forms.

The environment creates:

- Cognitive load
- Context switching
- Distraction
- Realistic interaction patterns



3. Phishing Simulation Flow

When the simulation starts:

- A session is created
- Multiple phishing emails are generated (AI or predefined)
- Emails contain:
 - Subtle domain mismatches
 - Legitimate-looking links
 - Fake links
 - Psychological pressure triggers (urgency, authority, reward)

The user may:

- Open the email
- Inspect sender
- Click legitimate link
- Click fake link
- Enter credentials
- Report phishing

The simulation does NOT end automatically.

The user manually stops the session.

This allows multiple behavioral signals to accumulate.

4. Behavioral Tracking Engine

Every interaction generates a structured event.

Example events:

- EMAIL_OPENED
- INSPECT_SENDER
- FAKE_LINK_CLICKED
- CREDENTIALS_ENTERED
- REPORT_PHISHING

The system stores:

- Event type
- Timestamp
- Order of actions

This allows sequence-level analysis.

We are not just checking correctness.

We are analyzing decision patterns.

5. Risk Evaluation Engine (Deterministic)

At session end:

The system evaluates the highest-risk behavior observed.

Example logic:

- CREDENTIALS_ENTERED → VERY HIGH
- FAKE_LINK_CLICKED → HIGH
- LEGIT_LINK_CLICKED → MEDIUM
- REPORT_PHISHING → LOW

This is rule-based.

It is:

- Explainable
- Transparent
- Defendable in viva

AI does NOT calculate risk.



6. Analytics Dashboard

The dashboard aggregates session data and provides:

- Total simulations completed
- Risk distribution
- Average vulnerability score
- Behavioral pattern counts
- Most recent session outcome

Over time, this creates a behavioral profile.

Example:

User frequently clicks urgency-based links but never enters credentials.

This is far more meaningful than a simple pass/fail score.

Where AI Is Used

AI is used in two places only:

1. Content generation
 - Generate realistic phishing email variations
 - Vary wording and psychological tone
2. Explanation generation
 - Convert structured risk result into human-readable feedback

AI never:

- Assigns risk
- Controls logic
- Alters scoring
- Changes rules

This keeps the system academically safe.



What Makes This Project Unique

This is NOT:

- A phishing quiz
- A basic training app
- A static simulator

This is:

A behavioral telemetry sandbox.

You are modeling:

- Hesitation patterns
- Action sequences
- Risk escalation
- Decision under pressure

You are turning human behavior into measurable signals.

Final System Flow

User logs in

- Starts simulation
- Desktop environment loads
- Phishing emails appear
- User interacts
- Events recorded
- User stops session
- Risk calculated
- AI explanation generated
- Dashboard updated

Complete loop.

Academic Value

This project touches:

- Cybersecurity awareness
- Human factors in security

- Behavioral analytics
- Gamification principles
- Explainable AI
- Risk modeling

It is interdisciplinary and strong.

What We Are NOT Building

We are NOT building:

- A real attack framework
- A penetration tool
- A malicious phishing system
- A reinforcement learning attacker
- A live adaptive adversary

We are building a defensive training and analysis platform.

In One Sentence

The Human Error Simulator is a gamified behavioral analytics platform that measures human susceptibility to social engineering attacks inside a controlled desktop simulation environment and converts user interactions into explainable risk insights.